

Recovery-Selection: Learning *Which* Recovery a Robot Policy Should Run

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Thesis. When a learned manipulation policy fails, the best way to recover depends on *what kind* of failure occurred, yet every current system hard-codes a single recovery behavior. I propose to make recovery-strategy *selection* a learnable, cost-aware decision problem, and to show that a small learned selector chooses recoveries that beat any fixed single-mechanism baseline at matched intervention budget.

1. The gap

Recovery from execution failures in robot learning is splintered into siloed lines that each commit to *one* mechanism: retry/rewind (RaC, SPR), local-RL recovery (RecoveryChaining), VLM re-planning (FailSafe, SC-VLA), and human-in-the-loop hand-off (the Cornell HITL framework, HRI 2026). Each works only in the regime it assumes (a rewind needs an in-distribution target; local RL needs a reachable funnel; VLM re-planning fails on perception-OOD states). No published work (verified over four substance-phrased literature probes during a structured search) does all of: (a) formally defines a policy’s *competence region* and types failures by their relation to it; (b) learns a meta-policy that selects among three or more heterogeneous recovery mechanisms under explicit cost and risk constraints; and (c) evaluates on real-robot rollouts with monolithic end-to-end VLA policies. The nearest neighbor (Cornell HITL) arbitrates only act-versus-ask and only for modular perception/planning/control pipelines, not monolithic VLAs.

2. Contribution

Three firsts, each independently checkable:

1. **A competence-grounded failure taxonomy.** Failure type is defined by the state’s relation to the frozen policy’s competence region (a calibrated combination of latent-state density and action-head ensemble disagreement, with a control-invariant safe set as the hard anchor), not by semantic error labels. The type implies the action: in-region $\rightarrow$ retry; near-boundary $\rightarrow$ rewind/back-off; out-of-region-plannable $\rightarrow$ re-plan; out-of-region-risky $\rightarrow$ ask-human.
2. **A learned meta-policy over heterogeneous recoveries.** A cost-sensitive selector maps (failure features, competence signals, task context) to one of five recovery arms, with a Lagrangian budget on the ask-human arm, unifying four currently siloed literatures.
3. **Counterfactual recovery evaluation.** For each injected failure we reset and replay the *entire* five-arm strategy grid to obtain per-arm cost (time, human-intervention effort, safety violations). This yields a per-failure oracle and a new metric, *recovery-regret* (realized cost minus per-failure oracle cost), replacing binary recovery-success.

3. Method and plan (simulation first, then UR3e)

- **Weeks 1–4 (sim, no robot) – the make-or-break gate.** Build the competence model on a frozen VLA (pi0-FAST or OpenVLA, inference/LoRA only). Inject 100–150 scripted failures across 3–4 tabletop tasks; replay the full five-arm grid per failure. **GO** if the per-failure oracle

beats the best fixed strategy by $\geq 15\%$ recovery-regret and a linear probe separates failure classes above $\sim 70\%$. **NO-GO** still yields a publishable benchmark plus metric.

- **Weeks 4–10.** Train and conformally calibrate the selector (offline contextual bandit, doubly-robust over the grid). Target ~ 400 – 600 labeled failure instances.
- **Weeks 10–16 (UR3e, with Lorena).** Validate recovery-regret against the four fixed-mechanism incumbents plus random and a hand-coded threshold rule; held-out task generalization; ask-rate/workload curves; ablations on each competence signal.

4. Feasibility and resources

No policy training; only a small selector head and competence calibration. Fits Tier-1 compute (1x H200). The simulation pilot doubles as the training dataset, capping downside at four weeks. The one robot-coupled stage (Weeks 10–16) needs scripted-failure UR3e time with repeatable resets, coordinated through Lorena per lab protocol.

5. Venue, alignment, and the ask

Venue. CoRL 2027 (primary), with an ICRA-workshop on-ramp; the dataset and recovery-regret metric are defensible contributions even if a selector appears elsewhere first. **Alignment.** The ask-human arm’s intervention-budget objective is a direct line into Lorena’s HRI / intervention-cost program, and the work reads foundational to ML committees while reading robotics to robotics committees, supporting the Fall-2027 PhD application. **The ask:** (1) access to the UR3e with repeatable resets for the Weeks 10–16 validation; (2) Lorena’s guidance on the human-intervention cost model (emulated for the main paper to avoid IRB timeline risk); (3) Prof. Padiar’s feedback on scope and venue framing.

Key related work (citations to finalize): Back-to-the-Manifold (competence/state-density); TAIL-Safe (control-invariant safe set); RaC, SPR (retry/rewind); RecoveryChaining (local-RL recovery); FailSafe, SC-VLA (VLM re-planning); Banerjee et al., Cornell HITL (HRI 2026); TCoT (AAAI 2026); SO-101 (recovery-aware evaluation).